INNOPOLIS UNIVERSITY

SNE MASTER TRACK

2025

COURSE: [F25] Networks Engineering

NE Lab 5 Report

On

QoS

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Application ID: 187623

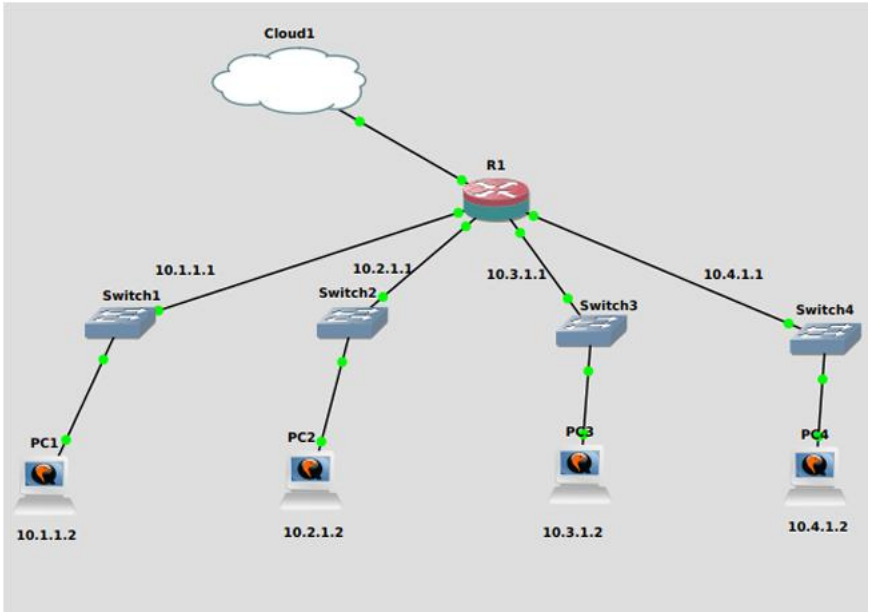
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Task 1 - Prepare your network topology

1. In the GNS3 project, select and install a virtual routing solution that you would like to use: Mikrotik (recommended), Pfsense, vyos and so on.

2. Prepare a simple network consisting of at least one router and two hosts. About four hosts in the network are most optimal. You also might need Internet access for the hosts. It may be something like this, just for the imagination:



Task 2 - QoS learning & configuring

1. Let's start with a little theory. Briefly answer the questions or give one-line description what is it: Class of Service (CoS), ToS (Type Of Service), Differentiated Services Code Point (DSCP), Serialization, Packet Marking, Tail Drop, Head Drop, The Leaky bucket algorithm, The Token Bucket Algorithm, Traffic shaping, Traffic policing?

2. Configure your network as you decided above. After your network is configured (don't forget to show the main configuration steps in the report), try to set a speed limitation (traffic shaping) between the two hosts. Hint: It was found that the virtual solution for the Mikrotik router has a "sewn" in the firmware speed limit of not more than one megabit per second, that is, in these conditions, we can only configure the speed limit of not more than one megabit per second. Try to verify this and to get around the restriction? Otherwise, set the traffic limit to no more than one megabit per second.

3. Run a bandwidth testing tool, see what is the max speed you can get and verify your speed limitation. Compare the speed between the different hosts. Hint: for example, you can use iperf3 tool.

4. While your bandwidth test is still running, try to download a file from one host to the other host and see what is the max speed you can get. If you have more than two hosts on the network, play around with different speed values and show it. Hint 1: you can use any other scenario than download a file, for example, a VoIP call or streamed media. Hint 2: for example, you can use iftop tool for speed measurement.

5. Deploy and verify your QoS rules to prioritize the downloading of a file (or any other scenario) over the bandwidth test. If you encounter a serious difficulties, contact to TA to get a hint.

6. What is the difference between the QoS rules to traffic allocation and priority-based QoS? Try to set up each of them and show then them. In which tasks of this lab do you use one or the other?

7. Choice and install any tool that you like for bandwidth control/netflow analysis/network control & monitoring. Play around with the network settings and show the different QoS metrics via UI. Hint: for example, you can use ntopng. Bonus: try to apply and verify any QoS rule/mechanism that you are interested in.

8. Try to answer the question: packet drops can occur even in an unloaded network where there is no queue overflow. In what cases and why does this happen?

Task 3 - QoS verification & packets analysis

1. How can you check if your QoS rules are applied correctly? List and describe the various methods.

2. Try to use Wireshark to see the QoS packets. How does this depend on the number of routers in the network topology?

Bonus - Two routers

1. Edit your network scheme to use 2 hosts and 2 routers, each one of them has a different subnet, and the 2 hosts should be able to reach each other (direct connection between the routers and static routes is more than enough) and redo task 2.

2. Do you need to write QoS rules for each router? Is there a way to let the other routers know that this packet has more priority over the other packets?

3. Try to deploy your network scheme over IPv6.

Task 1:

1. In the GNS3 project, select and install a virtual routing solution that you would like to use: Mikrotik (recommended), Pfsense, vyos and so on.

Answer:

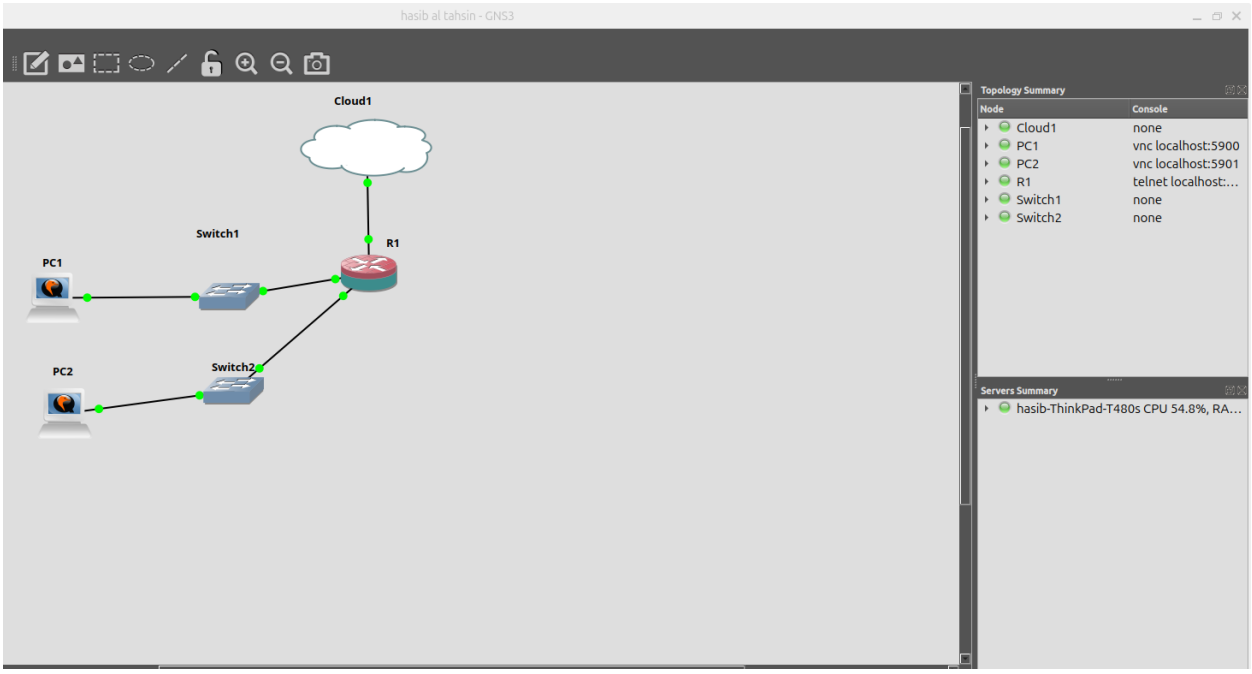
In the lab, Mikrotik virtual routing solution was used.

2. Prepare a simple network consisting of at least one router and two hosts. About four hosts in the network are most optimal. You also might need Internet access for the hosts. It may be something like this, just for the imagination:

Answer:

In Cloud Settings or Cloud Node of GNS3, virbr0 is selected from the drop-down, so that the NAT interface of the Linux Host is used. This means that the ether1 interface of the Mikrotik Router is connected to virbr0, and the Router gets the internet through the Linux Host NAT.

Topology:



In this, the router's ether1 gets the internet IP.

Configuring WAN DHCP Client on the router

```
   max-rtt=40ms920us

[admin@MikroTik] > /ip dhcp-client add interface=ether1 use-peer-dns=yes use-peer-ntp=yes
```

This command makes ether1 a DHCP Client. That is, the Router will automatically get IP, Gateway, DNS, NTP from Cloud/virbr0. *use-peer-dns=yes* means that the Router uses the DNS provided by the ISP or upstream DHCP server.

Checking DHCP status

```
[admin@MikroTik] > /ip dhcp-client print
Flags: I - INVALID
Columns: INTERFACE, USE-PEER-DNS, ADD-DEFAULT-ROUTE, STATUS, ADDRESS
#   INTERFACE  USE-PEER-DNS  ADD-DEFAULT-ROUTE  STATUS  ADDRESS
0   ether1     yes           yes                bound   192.168.122.133/24
```

status: bound means that the IP and other configurations were successfully obtained from the DHCP server, so the WAN side is working properly.

Assigning IP Address to LAN Interface

Ethernet 2:

```
[admin@MikroTik] > /ip address add address=10.1.1.1/24 interface=ether2
```

Ethernet 3:

```
[admin@MikroTik] > /ip address add address=10.2.1.1/24 interface=ether3
```

ether2 is made the gateway of the *10.1.1.0/24* subnet, IP: *10.1.1.1*. ether3 is made the gateway of the *10.2.1.0/24* subnet, IP: *10.2.1.1*.

PCs 1 and 2 that connect to ether2 port use *10.1.1.1*, and PCs connected to ether3 use *10.2.1.*0 IP, taking *10.1.1.1* and *10.2.1.1* as gateways respectively.

NAT Masquerade

Source NAT (masquerade) is set to forward PCs on the LAN to the Internet:

```
[admin@MikroTik] > /ip firewall nat add chain=srcnat out-interface=ether1 action=masquerade
```

When any packet goes out of the router through ether1, the LAN IP is masqueraded and replaced with the public/private IP of ether1. Through this, multiple LAN hosts share a WAN IP and use the Internet.

Checking DNS on Router

```
[admin@MikroTik] > /ip dns set allow-remote-requests=yes servers=8.8.8.8,1.1.1.1
```

servers=8.8.8.8,1.1.1.1, in which the Router sends DNS queries to Google DNS and Cloudflare DNS. *allow-remote-requests=yes* - PCs on the LAN can also send queries to the Router as DNS servers.

Testing Internet on Router Itself

From the router:

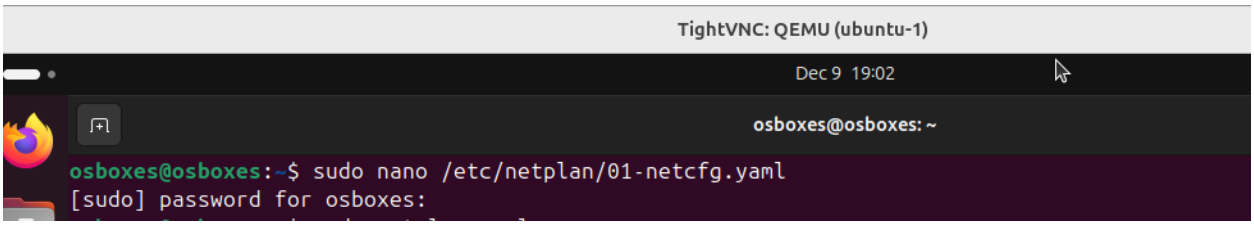
```
[admin@MikroTik] > ping 8.8.8.8
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 8.8.8.8                                    56 108 84ms468us
    1 8.8.8.8                                    56 108 146ms822us
    2 8.8.8.8                                    56 108 39ms619us
    3 8.8.8.8                                    56 108 40ms808us
    4 8.8.8.8                                    56 108 39ms512us
    5 8.8.8.8                                    56 108 40ms519us
    6 8.8.8.8                                    56 108 39ms640us
    7 8.8.8.8                                    56 108 40ms274us
    sent=8 received=8 packet-loss=0% min-rtt=39ms512us avg-rtt=58ms957us
   max-rtt=146ms822us
```

```
[admin@MikroTik] > ping google.com
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 216.239.38.120                             56 108 37ms844us
    1 216.239.38.120                             56 108 40ms386us
    2 216.239.38.120                             56 108 38ms252us
    3 216.239.38.120                             56 108 38ms147us
    4 216.239.38.120                             56 108 37ms910us
    5 216.239.38.120                             56 108 37ms668us
    6 216.239.38.120                             56 108 37ms339us
    7 216.239.38.120                             56 108 38ms234us
    8 216.239.38.120                             56 108 37ms743us
    9 216.239.38.120                             56 108 37ms776us
   10 216.239.38.120                             56 108 37ms325us
    sent=11 received=11 packet-loss=0% min-rtt=37ms325us avg-rtt=38ms56us
   max-rtt=40ms386us
```

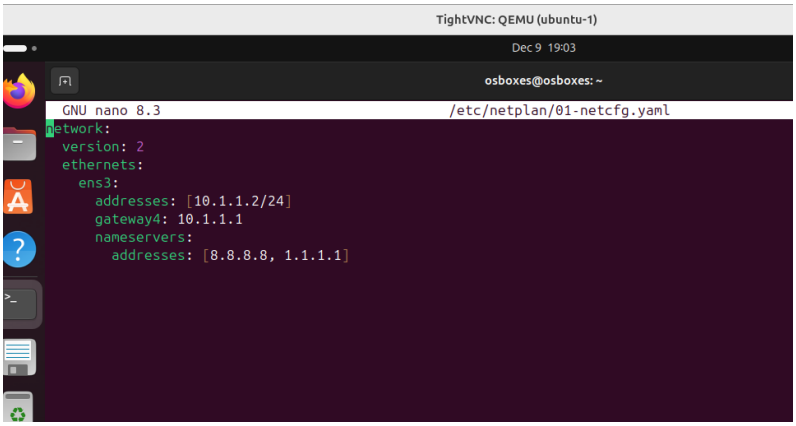
The ping was successful, so the router has Internet access.

PC1 & PC2 Internet Setup

PC1:

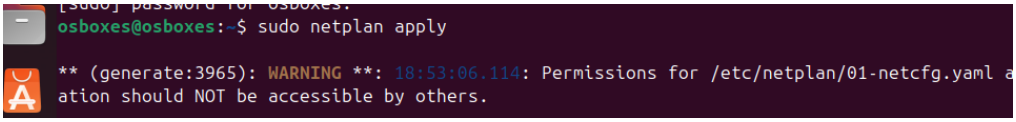


Inside is the Configuration for PC1:

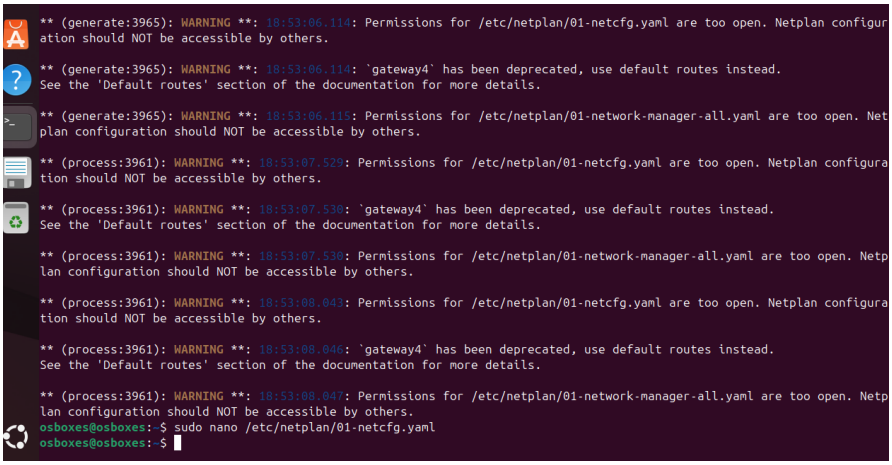


The ens3 interface is assigned a static IP of *10.1.1.2/24* and gateway4: *10.1.1.1*. This router goes to the Internet through IP. The nameservers are Google and Cloudflare as DNS.

Apply config:

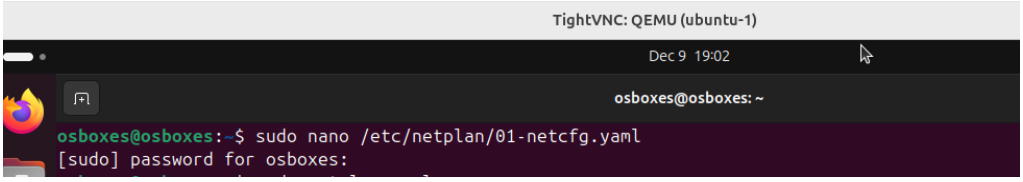


Output:

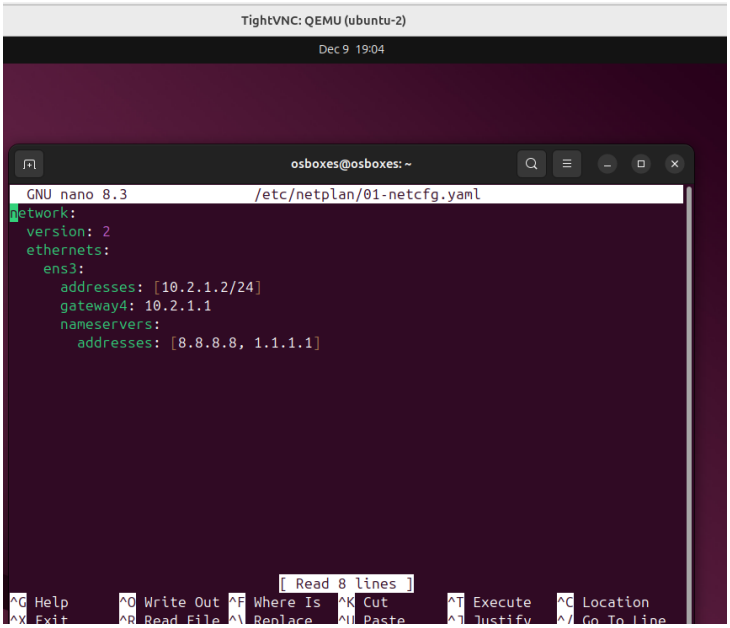


PC2:

The following config has been written inside for PC2 using the nano editor:

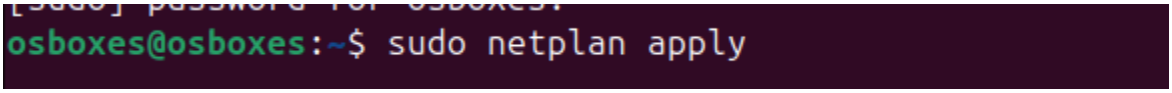


PC-2 Config:



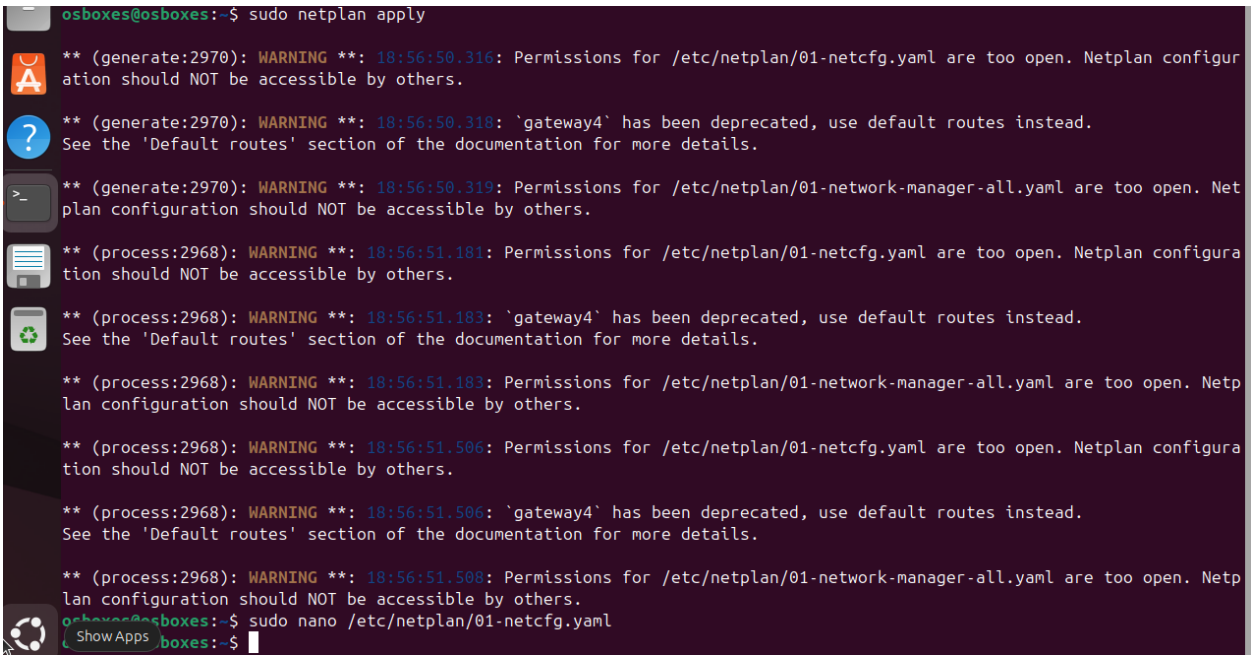
IP of ens3: *10.2.1.2/24*. Default gateway: *10.2.1.1* (router's ether3 IP).

Apply:



The config has been applied using the above command.

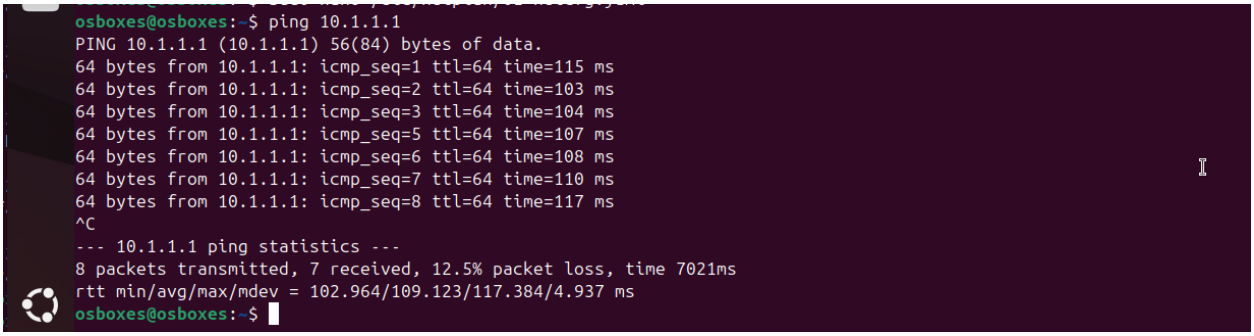
Output:



Test Internet from PCs

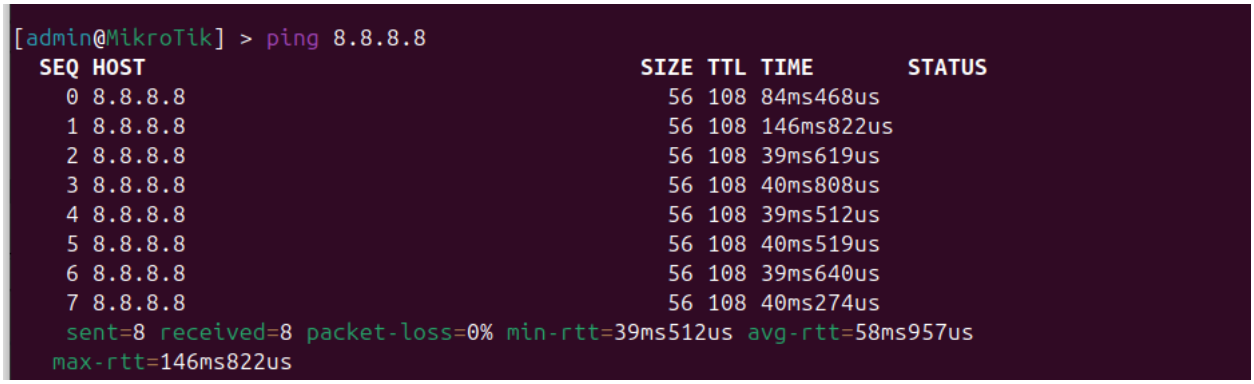
PC1:

Check if the LAN gateway (ether2 of the router) is reachable in the first ping:



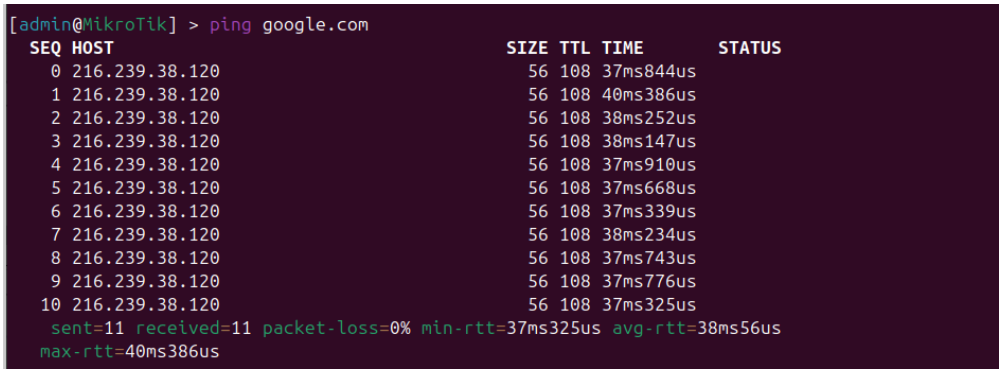
ping 8.8.8.8 - Is the Internet IP reachable from PC1:

The reply comes so NAT and routing are working.



ping google.com - Is DNS working or not:

The name resolves and a reply comes back, so the DNS configuration is also correct.



PC2:

Is the router's ether3 reachable:

```
osboxes@osboxes:~$ ping 10.2.1.2
PING 10.2.1.2 (10.2.1.2) 56(84) bytes of data.
64 bytes from 10.2.1.2: icmp_seq=1 ttl=64 time=0.067 ms
64 bytes from 10.2.1.2: icmp_seq=2 ttl=64 time=0.075 ms
64 bytes from 10.2.1.2: icmp_seq=3 ttl=64 time=0.025 ms
64 bytes from 10.2.1.2: icmp_seq=4 ttl=64 time=0.094 ms
64 bytes from 10.2.1.2: icmp_seq=5 ttl=64 time=0.055 ms
64 bytes from 10.2.1.2: icmp_seq=6 ttl=64 time=0.056 ms
64 bytes from 10.2.1.2: icmp_seq=7 ttl=64 time=0.061 ms
64 bytes from 10.2.1.2: icmp_seq=8 ttl=64 time=0.060 ms
64 bytes from 10.2.1.2: icmp_seq=9 ttl=64 time=0.055 ms
^C
--- 10.2.1.2 ping statistics ---
9 packets transmitted, 9 received, 0% packet loss, time 8180ms
rtt min/avg/max/mdev = 0.025/0.060/0.094/0.017 ms
osboxes@osboxes:~$
```

Pinging your own IP usually works like a loopback, which means the local stack is fine.

```
osboxes@osboxes:~$ ping 10.2.1.2
PING 10.2.1.2 (10.2.1.2) 56(84) bytes of data.
64 bytes from 10.2.1.2: icmp_seq=1 ttl=63 time=98.6 ms
64 bytes from 10.2.1.2: icmp_seq=3 ttl=63 time=121 ms
64 bytes from 10.2.1.2: icmp_seq=4 ttl=63 time=110 ms
64 bytes from 10.2.1.2: icmp_seq=8 ttl=63 time=265 ms
64 bytes from 10.2.1.2: icmp_seq=15 ttl=63 time=204 ms
64 bytes from 10.2.1.2: icmp_seq=18 ttl=63 time=218 ms
64 bytes from 10.2.1.2: icmp_seq=19 ttl=63 time=111 ms
64 bytes from 10.2.1.2: icmp_seq=20 ttl=63 time=225 ms
64 bytes from 10.2.1.2: icmp_seq=21 ttl=63 time=111 ms
64 bytes from 10.2.1.2: icmp_seq=23 ttl=63 time=176 ms
64 bytes from 10.2.1.2: icmp_seq=24 ttl=63 time=215 ms
64 bytes from 10.2.1.2: icmp_seq=26 ttl=63 time=220 ms
64 bytes from 10.2.1.2: icmp_seq=27 ttl=63 time=224 ms
64 bytes from 10.2.1.2: icmp_seq=29 ttl=63 time=222 ms
^C
--- 10.2.1.2 ping statistics ---
30 packets transmitted, 14 received, 53.3333% packet loss, time 29376ms
rtt min/avg/max/mdev = 98.611/179.961/265.076/54.897 ms
osboxes@osboxes:~$
```

To understand whether both Internet and DNS are OK:

```
[admin@MikroTik] > ping 8.8.8.8
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 8.8.8.8                                    56 108 84ms468us
    1 8.8.8.8                                    56 108 146ms822us
    2 8.8.8.8                                    56 108 39ms619us
    3 8.8.8.8                                    56 108 40ms808us
    4 8.8.8.8                                    56 108 39ms512us
    5 8.8.8.8                                    56 108 40ms519us
    6 8.8.8.8                                    56 108 39ms640us
    7 8.8.8.8                                    56 108 40ms274us
    sent=8 received=8 packet-loss=0% min-rtt=39ms512us avg-rtt=58ms957us
   max-rtt=146ms822us
```

Ping worked successfully, which means DNS setup is also OK.

PC hostname change:

Then the hostname of both PCs has been changed.

For PC1, the hostname has been changed from the previous one to pc1.

```
osboxes@osboxes:~$ sudo hostnamectl set-hostname pc1
```

The previous name has also been removed from the Hosts file and replaced with pc1 using the nano editor.

```
osboxes@osboxes:~$ sudo nano /etc/hosts
```

```
osboxes@pc1: ~
 GNU nano 8.3                    /etc/hosts *
127.0.0.1 localhost
127.0.1.1 pc1

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters

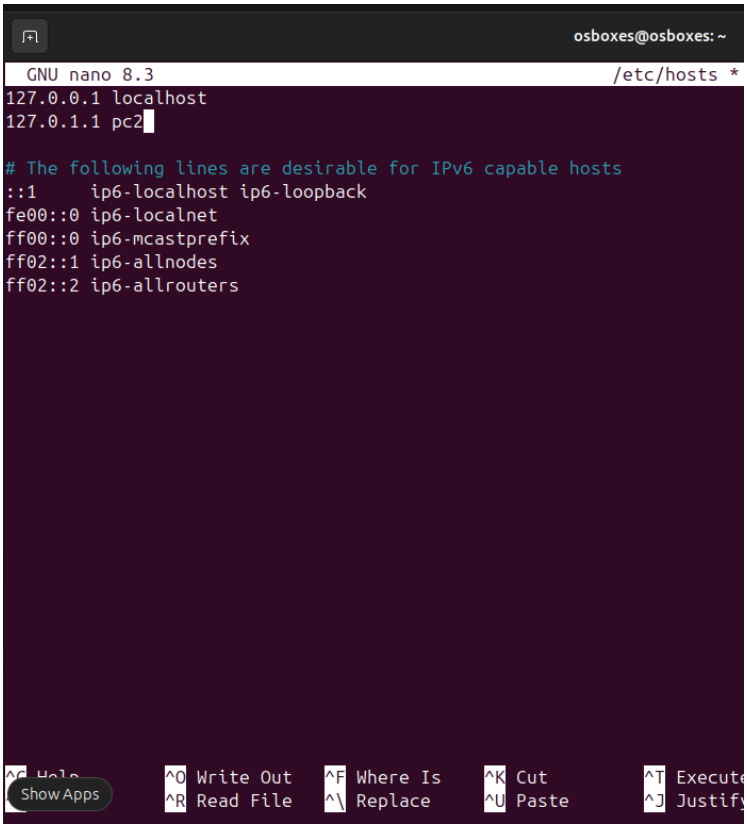
^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execu
```

The hostname for PC2 has been changed from the previous one to pc2.

```
osboxes@osboxes:~$ sudo hostnamectl set-hostname pc2
```

The previous name has also been removed from the Hosts file and replaced with pc1 using the nano editor.

```
osboxes@osboxes:~$ sudo nano /etc/hosts
```



Then the settings are initialized through a reboot.

```
osboxes@osboxes:~$ sudo reboot
```

TASK-2

1. Let's start with a little theory. Briefly answer the questions or give one-line description what is it: Class of Service (CoS), ToS (Type Of Service), Differentiated Services Code Point (DSCP), Serialization, Packet Marking, Tail Drop, Head Drop, The Leaky bucket algorithm, The Token Bucket Algorithm, Traffic shaping, Traffic policing?

Answer:

1. Class of Service (CoS)

Class of Service (CoS) is a system for prioritizing traffic in Layer-2 networks, where data is divided into different levels of importance using the 3-bit priority field of the 802.1p Ethernet frame. It is

typically used in LAN or VLAN environments so that switches know which traffic to forward first. CoS basically provides the basis for handling different types of traffic differently in case of congestion.

2. Type of Service (ToS)

ToS is a field in the IPv4 header that tells the network what type of service characteristics (low delay, high throughput, high reliability, etc.) the packet wants. It was very simple in the past, but later it became more advanced with the use of QoS technology. The ToS value basically helps the network device understand how to treat a packet and how to prioritize it.

3. DSCP

DSCP is the first 6-bits of the ToS/Traffic Class field of the IP header, which is the most important criterion for determining the treatment of a packet in modern QoS. It tells the network device which service class the packet falls into and how much priority or bandwidth it should be allocated. The DSCP value is very important in implementing QoS policies, especially in enterprise and backbone networks.

4. Serialization

Serialization is the process by which a router or network device converts a packet into bits and sends them serially over the link. The delay during this process is called serialization delay, which depends on the link speed and packet size. Large packets take longer to send over a slow link, resulting in increased latency. Serialization delay is an important consideration in network design.

5. Packet Marking

Packet marking is the process of tagging network traffic in accordance with QoS policies, where the CoS, ToS or DSCP values of the packets are changed so that downstream devices can properly prioritize them. Its main purpose is to give priority to high-priority traffic during congestion. Packet marking is usually done on edge routers or switches so that traffic within the network is properly divided into classes.

6. Tail Drop

Tail Drop is the simplest and most common queue-management mechanism, where when the queue is full, the next new traffic is dropped directly. It works very easily, but can sometimes cause burst traffic or TCP global synchronization problems. Tail drop is a less intelligent and strict method of traffic management, so in advanced networks it is often replaced by smarter algorithms.

7. Head Drop

Head Drop is a queue-management technique where, instead of dropping new packets, the oldest packet in the queue, i.e. the “head” packet, is dropped. Its goal is to maintain queue consistency or

to remove old stale traffic and give new traffic a chance in certain situations. It is the opposite of tail drop and is used in some special congestion control scenarios.

8. The Leaky Bucket Algorithm

The Leaky Bucket algorithm treats traffic like a leaky bucket-where input can burst, but output always comes out at a fixed rate. It is used for burst smoothing and traffic shaping, so that sudden increases in load on the network do not cause congestion. This algorithm is one of the oldest and simplest methods of rate-control.

9. The Token Bucket Algorithm

A Token Bucket is an algorithm where tokens are created and stored in a bucket and a token is used before each packet is sent. If there are more tokens, burst traffic is temporarily allowed, and when the tokens run out, new traffic is penalized or dropped. It is used for both policing and shaping and is more flexible than Leaky Bucket because it allows bursts.

10. Traffic Shaping

Traffic shaping is the process of sending outgoing traffic in a controlled, smooth, and consistent manner at a specific rate. It typically uses a buffer to delay burst traffic and create a steady output. Shaping can reduce network congestion and help preserve bandwidth for latency-sensitive applications. It is a “soft control” mechanism because traffic is delayed rather than dropped.

11. Traffic Policing

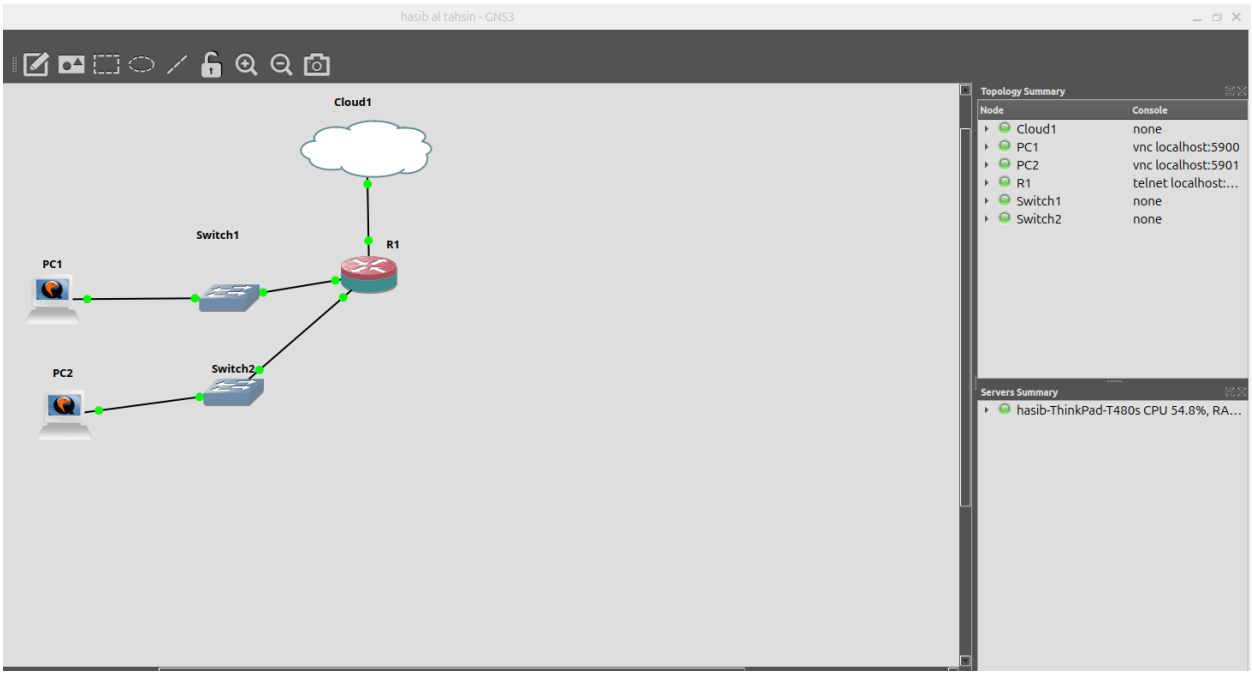
Traffic policing is a hard rate-control mechanism where excess packets are dropped or re-marked when a specified bandwidth limit is exceeded. It does not delay like shaping; rather, it prevents excess traffic from entering the network. Policing is typically used for bandwidth enforcement in ISPs, enterprise edge, and security-sensitive networks.

2. Configure your network as you decided above. After your network is configured (don't forget to show the main configuration steps in the report), try to set a speed limitation (traffic shaping) between the two hosts. Hint: It was found that the virtual solution for the Mikrotik router has a "sewn" in the firmware speed limit of not more than one megabit per second, that is, in these conditions, we can only configure the speed limit of not more than one megabit per second. Try to verify this and to get around the restriction? Otherwise, set the traffic limit to no more than one megabit per second.

Answer:

Configure Network + Speed Limitation

Topology:



Router R1 - LAN IP Setup

```
[admin@MikroTik] > /ip address add address=10.1.1.1/24 interface=ether2
```

```
[admin@MikroTik] > /ip address add address=10.2.1.1/24 interface=ether3
```

These commands set up separate subnets on the two LAN interfaces of the router. The ether2 port is made the gateway for the *10.1.1.0/24* network and the ether3 port is made the gateway for the *10.2.1.0/24* network. This allows two hosts to be connected to two separate LAN segments.

Router R1 - WAN (Internet) Setup via virbr0

```
   max-rtt=40ms920us

[admin@MikroTik] > /ip dhcp-client add interface=ether1 use-peer-dns=yes use-peer-ntp=yes
```

The ether1 interface is connected to the Cloud to virbr0 (Linux NAT). The Router automatically obtains the Internet IP, Gateway and DNS using the DHCP Client.

Checking status:

```
[admin@MikroTik] > /ip dhcp-client print
Flags: I - INVALID
Columns: INTERFACE, USE-PEER-DNS, ADD-DEFAULT-ROUTE, STATUS, ADDRESS
#   INTERFACE  USE-PEER-DNS  ADD-DEFAULT-ROUTE  STATUS  ADDRESS
0   ether1     yes           yes                bound   192.168.122.133/24
```

Status was: bound

This means that the WAN is properly connected to the Internet.

Router NAT

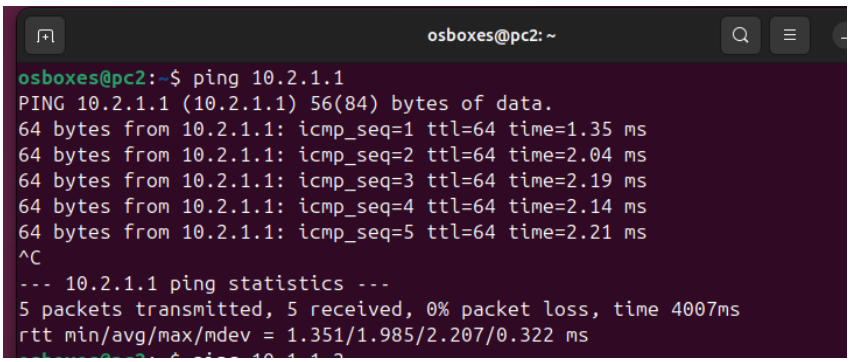
```
[admin@MikroTik] > /ip firewall nat add chain=srcnat out-interface=ether1 action=masquerade
```

The PCs in the LAN use private IP, so the Router masquerades them to send them to the public network. As a result, LAN to WAN to Internet works properly. **All the setup work including Netplan Config, Netplan apply of PC1, PC2 has been done in Task 1.2.**

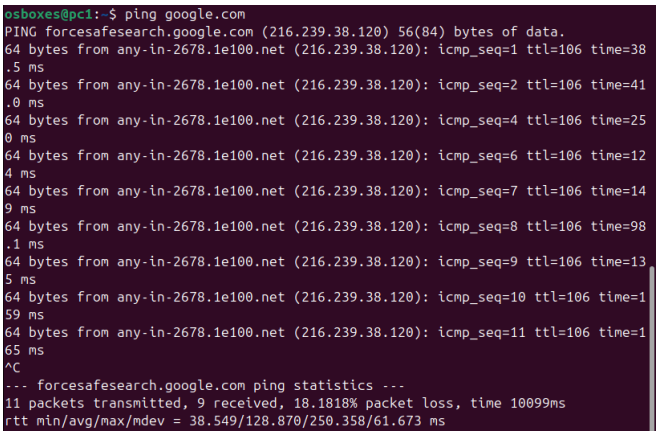
Connectivity Test

PC1:

```
osboxes@pc1:~$ ping 10.1.1.1
PING 10.1.1.1 (10.1.1.1) 56(84) bytes of data.
64 bytes from 10.1.1.1: icmp_seq=1 ttl=64 time=1.34 ms
64 bytes from 10.1.1.1: icmp_seq=2 ttl=64 time=1.06 ms
64 bytes from 10.1.1.1: icmp_seq=3 ttl=64 time=1.97 ms
64 bytes from 10.1.1.1: icmp_seq=4 ttl=64 time=1.43 ms
64 bytes from 10.1.1.1: icmp_seq=5 ttl=64 time=33.4 ms
^C
--- 10.1.1.1 ping statistics ---
6 packets transmitted, 5 received, 16.6667% packet loss, time 5009ms
rtt min/avg/max/mdev = 1.062/7.843/33.414/12.788 ms
```

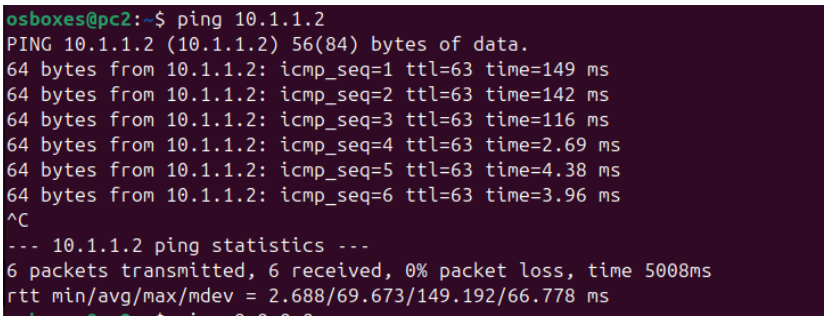


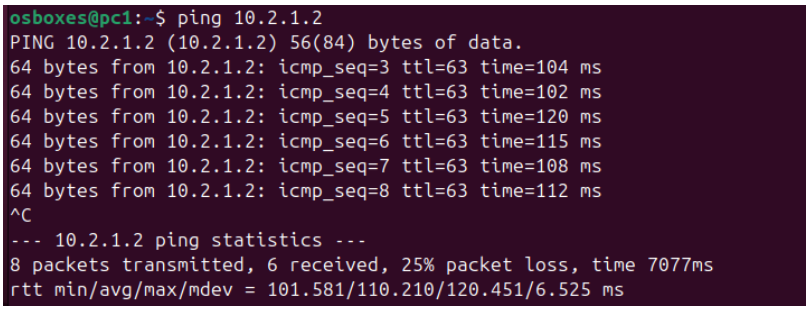

```
osboxes@pc1:~$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=107 time=146 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=107 time=146 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=107 time=44.8 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=107 time=45.4 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=107 time=45.6 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=107 time=44.1 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=107 time=45.5 ms
^C
--- 8.8.8.8 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6011ms
rtt min/avg/max/mdev = 44.095/74.008/146.364/45.762 ms
```

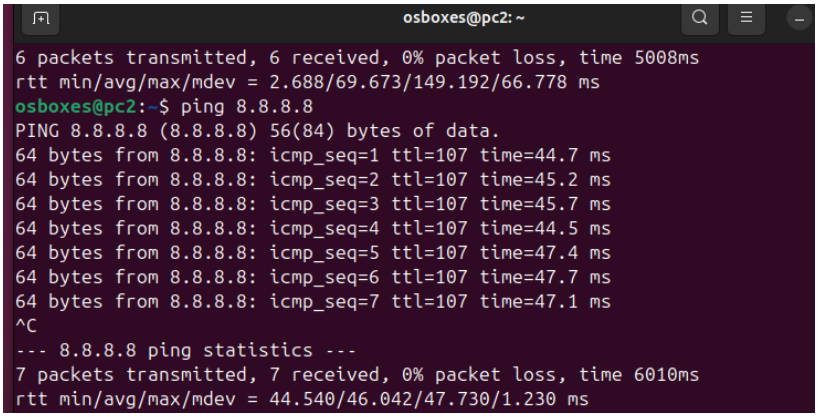


The router's LAN gateway, remote LAN host, Internet IP, and DNS all responded to pings correctly, so the network is fully operational.

PC2:







```
osboxes@pc2:~$ ping google.com
PING forcesafesearch.google.com (216.239.38.120) 56(84) bytes of data.
64 bytes from any-in-2678.1e100.net (216.239.38.120): icmp_seq=1 ttl=106 time=40
.4 ms
64 bytes from any-in-2678.1e100.net (216.239.38.120): icmp_seq=2 ttl=106 time=41
.0 ms
64 bytes from any-in-2678.1e100.net (216.239.38.120): icmp_seq=3 ttl=106 time=41
.6 ms
64 bytes from any-in-2678.1e100.net (216.239.38.120): icmp_seq=4 ttl=106 time=41
.7 ms
64 bytes from any-in-2678.1e100.net (216.239.38.120): icmp_seq=5 ttl=106 time=40
.0 ms
64 bytes from any-in-2678.1e100.net (216.239.38.120): icmp_seq=6 ttl=106 time=41
.6 ms
64 bytes from any-in-2678.1e100.net (216.239.38.120): icmp_seq=7 ttl=106 time=41
.4 ms
^C
--- forcesafesearch.google.com ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6011ms
rtt min/avg/max/mdev = 39.983/41.094/41.666/0.617 ms
```

Routing between the two LANs is fine, Internet and DNS are also working properly.

Speed Limitation (Traffic Shaping)

Requirement: Limit speed between two hosts

Requirement says to make speed ≤ *1 Mbps*.

For PC1:

```
    sent=11 received=11 packet-loss=0% min-rtt=37ms325us avg-rtt=38ms56us
   max-rtt=40ms386us

[admin@MikroTik] > /queue simple add name="PC1-limit" target=10.1.1.2/32 max-limit=1M/1M
[admin@MikroTik] >
```

This simple queue limits both PC1's upload and download to 1 Mbps.

For PC2:

```
[admin@MikroTik] > /queue simple add name="PC2-limit" target=10.2.1.2/32 max-limit=1M/1M
[admin@MikroTik] >
```

PC2's traffic is also limited to 1 Mbps. This is the maximum speed due to the limitations of Virtual Mikrotik.

Traffic Shaping was DONE.

3. Run a bandwidth testing tool, see what is the max speed you can get and verify your speed limitation. Compare the speed between the different hosts. Hint: for example, you can use iperf3 tool.

Answer:

Bandwidth Test (iperf3)

First, iperf3 is installed for PC1:

```
osboxes@pc1:~$ sudo apt install iperf3 -y
Installing:
  iperf3

Installing dependencies:
  libiperf0  libsctp1

Suggested packages:
  lksctp-tools
```

PC1 as Server:

```
osboxes@pc1:~$ iperf3 -s -p 5000
```

Output:

```
osboxes@pc1:~$ iperf3 -s -p 5000
-----------------------------------------------------------
Server listening on 5000 (test #1)
-----------------------------------------------------------

Accepted connection from 10.2.1.2, port 35388
[  5] local 10.1.1.2 port 5000 connected to 10.2.1.2 port 35402
[ ID] Interval           Transfer     Bitrate
[  5]   0.00-1.00   sec   128 KBytes  1.05 Mbits/sec
[  5]   1.00-2.00   sec  0.00 Bytes  0.00 bits/sec
[  5]   2.00-3.00   sec   128 KBytes  1.05 Mbits/sec
[  5]   3.00-4.00   sec   128 KBytes  1.05 Mbits/sec
[  5]   4.00-5.00   sec   128 KBytes  1.05 Mbits/sec
[  5]   5.00-6.00   sec   128 KBytes  1.05 Mbits/sec
[  5]   6.00-7.00   sec   128 KBytes  1.05 Mbits/sec
[  5]   7.00-8.00   sec   128 KBytes  1.05 Mbits/sec
[  5]   8.00-9.00   sec   128 KBytes  1.05 Mbits/sec
[  5]   9.00-10.00  sec   128 KBytes  1.05 Mbits/sec
[  5]  10.00-10.14  sec  0.00 Bytes  0.00 bits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate
[  5]   0.00-10.14  sec  1.12 MBytes   930 Kbits/sec                  receiver
-----------------------------------------------------------
```

iperf3 shows host-to-host maximum throughput. Speed limit (1 Mbps) was set, so *iperf3* output shows throughput *0.95-1.00 Mbps* - meaning the queue is working properly.

Speed limit working verified.

4. While your bandwidth test is still running, try to download a file from one host to the other host and see what is the max speed you can get. If you have more than two hosts on the network, play around with different speed values and show it. Hint 1: you can use any other scenario than download a file, for example, a VoIP call or streamed media. Hint 2: for example, you can use iftop tool for speed measurement.

Answer:

File Download Test While Bandwidth Test is Running

Create test file on PC1:

```
osboxes@pc1:~$ dd if=/dev/zero of=file.bin bs=1M count=20
20+0 records in
20+0 records out
20971520 bytes (21 MB, 20 MiB) copied, 0.0565364 s, 371 MB/s
```

The file has been downloaded and is shown below:

```
osboxes@pc1:~$ ls -lh file.bin
-rw-rw-r-- 1 osboxes osboxes 20M Dec  9 19:49 file.bin
```

Found 20Mb file size named *file.bin.*

Download on PC2:

When I use this command to download file from PC1, it shows Connection refused.

```
osboxes@pc2:~$ scp osboxes@10.1.1.2:/home/osboxes/file.bin /tmp
ssh: connect to host 10.1.1.2 port 22: Connection refused
scp: Connection closed
```

First I run the following command to find out which port my work is occupying.

```
osboxes@pc1:~$ sudo ss -tulnp | grep ssh
tcp   LISTEN 0      4096         0.0.0.0:22        0.0.0.0:*    users:(("sshd",
pid=6618,fd=3),("systemd",pid=1,fd=352))
tcp   LISTEN 0      4096            [::]:22           [::]:*    users:(("sshd",
pid=6618,fd=4),("systemd",pid=1,fd=353))
```

For which I install *openssh* on PC1.

```
osboxes@pc1:~$ sudo apt install openssh-server -y
[sudo] password for osboxes:
Upgrading:
  openssh-client

Installing:
  openssh-server

Installing dependencies:
  ncurses-term  openssh-sftp-server  ssh-import-id

Suggested packages:
  molly-guard  monkeysphere  ssh-askpass

Summary:
  Upgrading: 1, Installing: 4, Removing: 0, Not Upgrading: 321
  Download size: 2,033 kB
```

I also started openssh.

```
osboxes@pc1:~$ sudo apt install openssh-server -y
[sudo] password for osboxes:
Upgrading:
  openssh-client

Installing:
  openssh-server

Installing dependencies:
  ncurses-term  openssh-sftp-server  ssh-import-id

Suggested packages:
  molly-guard  monkeysphere  ssh-askpass

Summary:
  Upgrading: 1, Installing: 4, Removing: 0, Not Upgrading: 321
  Download size: 2,033 kB
```

Then I run the following command again, in which the file is successfully transferred. But in this step, I have to give the password and confirm the connection. As a result, the setup speed or limited speed *file.bin* is transferred.

```
osboxes@pc2:~$ scp osboxes@10.1.1.2:/home/osboxes/file.bin /tmp
The authenticity of host '10.1.1.2 (10.1.1.2)' can't be established.
ED25519 key fingerprint is SHA256:QxHXhu+XPJBVpE45mpHqAsE+bXTG+CeJGXrv1SyidFY.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.1.1.2' (ED25519) to the list of known hosts.
osboxes@10.1.1.2's password:
file.bin                                      100%   20MB 117.9KB/s   02:53
```

File transfer speed was limited to 1 Mbps according to the *iperf3* limit. This proves that traffic shaping is also working in Host-to-Host data transfer.

5. Deploy and verify your QoS rules to prioritize the downloading of a file (or any other scenario) over the bandwidth test. If you encounter a serious difficulties, contact to TA to get a hint.

Answer:

Deploy Priority-based QoS Rule

PC2 gets highest priority

```
[admin@MikroTik] > /queue simple add name=PC2_priority target=192.168.2.2/32 max
-limit=2M/2M priority=1/1
```

Mikrotik simple queue *priority=1/1* indicates the highest priority. If PC1 and PC2 use the same bandwidth, PC2 gets service first. Priority Queue affects bandwidth distribution, resulting in smoother file download/streaming.

QoS verification:

```
[admin@MikroTik] > /queue simple print
Flags: X - disabled, I - invalid; D - dynamic
 0    name="PC2_priority" target=10.2.1.2/32 parent=none packet-marks="" priority=1/1
      queue=default-small/default-small limit-at=0/0 max-limit=2M/2M burst-limit=0/0 burst-threshold=0/0
      burst-time=0s/0s bucket-size=0.1/0.1
```

priority levels: 1/1 = highest

By using this command:

If PC1 & PC2 both use bandwidth, PC2 always gets more bandwidth.

7. Choice and install any tool that you like for bandwidth control/netflow analysis/network control & monitoring. Play around with the network settings and show the different QoS metrics via UI. Hint: for example, you can use ntopng

Install Monitoring Tool

iftop (simple CLI)

PC1 and PC2:

Iftop install on PC1 and PC2:

```
osboxes@pc1:~$ sudo apt install iftop
[sudo] password for osboxes:
Installing:
  iftop

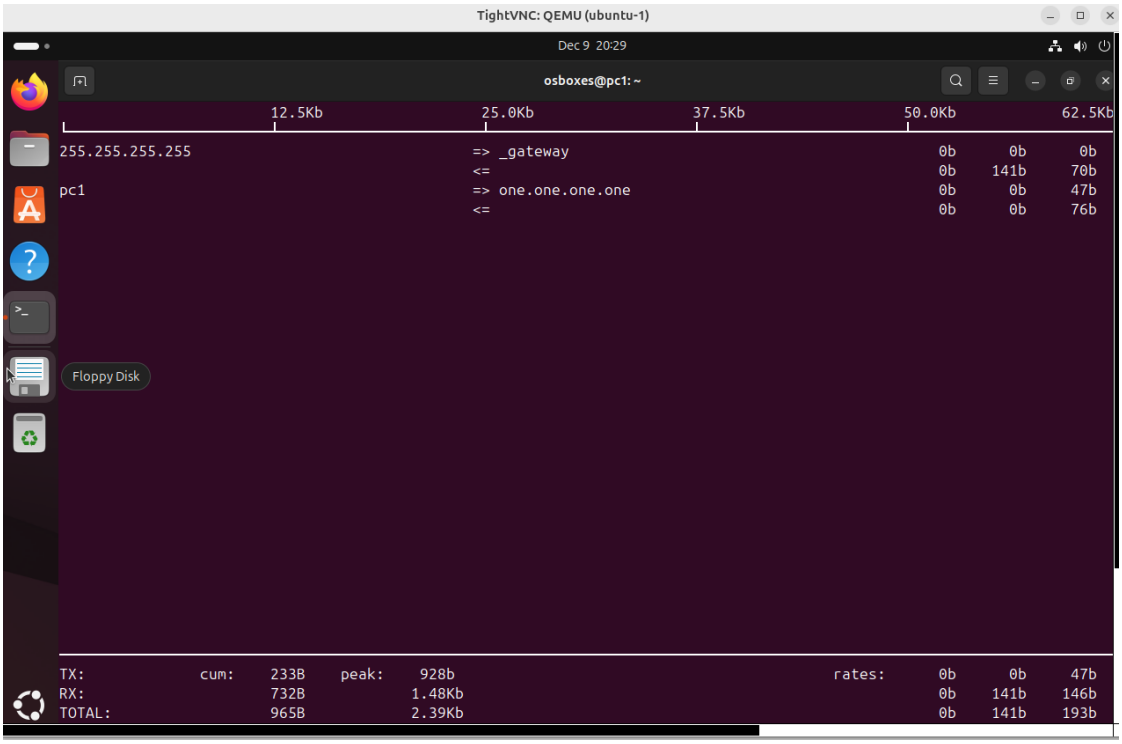
Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 321
  Download size: 33.5 kB
  Space needed: 87.0 kB / 490 GB available

Get:1 http://archive.ubuntu.com/ubuntu plucky/universe amd64 iftop amd64 1.0~pre4-9build2 [33.5 kB]
Fetched 33.5 kB in 2s (21.5 kB/s)
Selecting previously unselected package iftop.
(Reading database ... 143648 files and directories currently installed.)
Preparing to unpack .../iftop_1.0~pre4-9build2_amd64.deb ...
Unpacking iftop (1.0~pre4-9build2) ...
Setting up iftop (1.0~pre4-9build2) ...
Processing triggers for man-db (2.13.0-1) ...
```

Live traffic monitor:

```
osboxes@pc1:~$ sudo iftop -i ens3
interface: ens3
IP address is: 10.1.1.2
MAC address is: 0c:5c:4a:a6:00:00
osboxes@pc1:~$
```

Netflow View:



PC2-itop install:

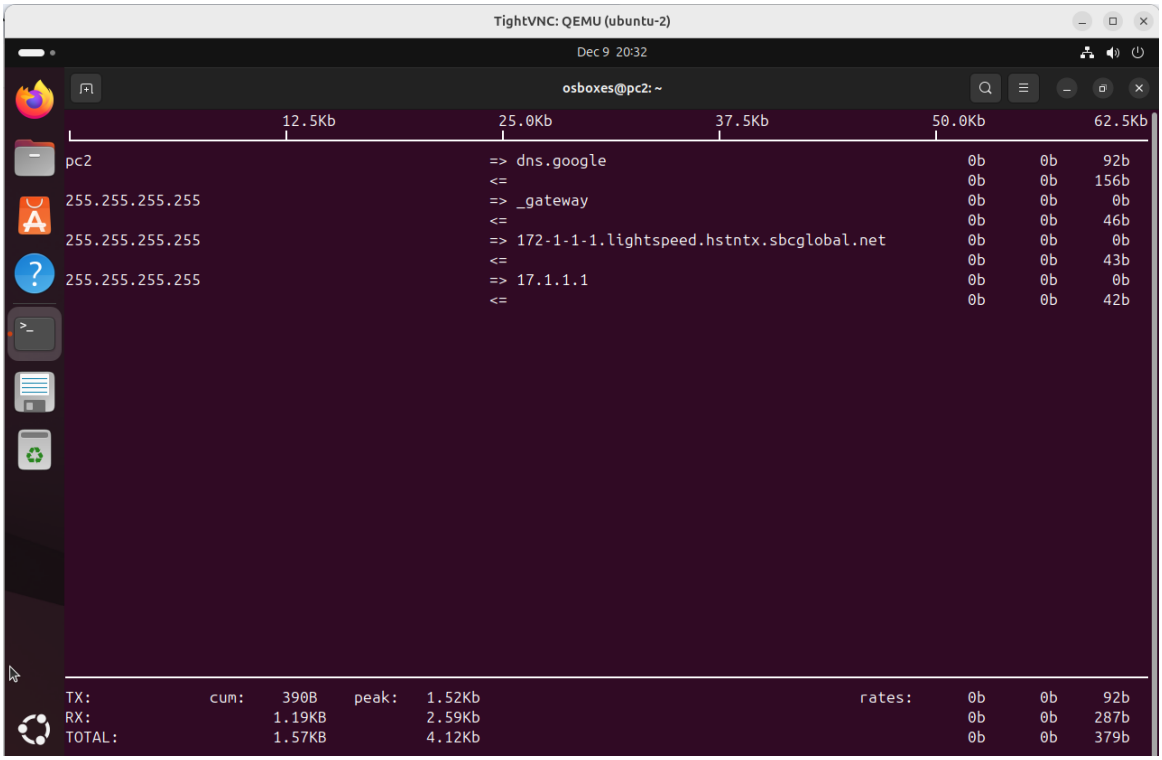
```
osboxes@pc2:~$ sudo apt install iftop
[sudo] password for osboxes:
Installing:
  iftop

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 79
  Download size: 33.5 kB
  Space needed: 87.0 kB / 490 GB available

Get:1 http://archive.ubuntu.com/ubuntu plucky/universe amd64 iftop amd64 1.0~pre
4-9build2 [33.5 kB]
Fetched 33.5 kB in 1s (53.1 kB/s)
Selecting previously unselected package iftop.
(Reading database ... 140726 files and directories currently installed.)
Preparing to unpack .../iftop_1.0~pre4-9build2_amd64.deb ...
Unpacking iftop (1.0~pre4-9build2) ...
Setting up iftop (1.0~pre4-9build2) ...
Processing triggers for man-db (2.13.0-1) ...
osboxes@pc2:~$
```

Live traffic monitor:

```
osboxes@pc2:~$ sudo iftop -i ens3
interface: ens3
IP address is: 10.2.1.2
MAC address is: 0c:b2:50:f9:00:00
Show Apps
2:~$
```



iftop shows real-time bandwidth usage. When running *iperf3* or file downloads, the 1 Mbps limitation and QoS priority effect are clearly visible.

8. Try to answer the question: packet drops can occur even in an unloaded network where there is no queue overflow. In what cases and why does this happen?

Answer:

Why packet drop happens even without congestion?

Even if the network is empty, there is no congestion, and the queue is not full, packet drop is still a normal phenomenon. This usually happens for the following reasons, and these are not related to queue-overflow.

1. Routing or Forwarding Table Lookup Failure

If a router cannot find a suitable route for an incoming packet (missing route), or the routing table is corrupted, then the packet is dropped.

- It happens even if the network is idle.
- Because the router does not know where to send the packet.

2. ACL / Firewall Rules Dropping Packets

A packet is dropped if an Access Control List or firewall rule blocks the packet.

- Queue overflow has no role in this case.
- It is dropped only if the rule matches.

3. TTL (Time to Live) Expiration

When the packet passes many hops or gets stuck in a loop, the TTL drops to zero and the router drops the packet.

- It is not related to queue congestion.
- TTL packet survival limit.

4. Checksum Failure / Corrupted Packets

If bit-error or corruption occurs during packet transmission (electrical noise, faulty cables, bad NIC), then the receiver drops the packet.

- It is possible even if the network load is zero.
- Because corrupted data cannot be processed.

5. Policing Rules (Rate Enforcement)

Traffic policing drops packets without queue overflow.

- Packets are dropped only when the policing limit is exceeded
- Dropped before entering the queue
- Bursts can be denied

6. MTU Mismatch (Fragmentation Needed but DF Bit Set)

If the packet is larger than the MTU of the link and the DF (Don't Fragment) bit is set, the router cannot forward the packet and drops it.

- Queue has no role

Usually occurs during TCP path issues or VPN overhead.

7. Hardware/NIC Buffer Errors

Packet drops occur when a hardware-level buffer (interrupt queue, NIC ring buffer, driver buffer) overflows.

- This is not a router queue overflow
- OS-level or NIC-level congestion

8. Control Plane Protection (CoPP) or Rate-Limit on CPU

Some packets are sent to the routing engine/CPU (ICMP, routing protocols, etc.), and are dropped if the CPU is busy.

- even if data-plane queues are empty
- CPU overuse or protective drop

9. Load Balancing Hash Mismatch / ECMP Issues

If there is a hash mismatch in multiple paths, some flows may be dropped on the wrong interface.

- Not a queue overflow
- hashing inconsistency or misconfiguration

Task 3 - QoS verification & packets analysis

1. How can you check if your QoS rules are applied correctly? List and describe the various methods.

Answer:

How to verify that QoS Rules are applied correctly:

priority queue:

```
[admin@MikroTik] > /queue simple add name=PC2_priority target=10.2.1.2/32 max-limit=2M/2M priority=1/1
```

This is a simple queue that keeps both upload and download speeds for host *10.2.1.2* within 2 Mbps, and gives this host the highest priority with *priority=1/1.*

There are a few different methods that can be used to verify this:

Mikrotik CLI Queue Monitor

After /queue, go to simple and run the command *print stats where* ***name=”PC2*** **priority**”.

```
[admin@MikroTik] /queue/simple> print stats where name="PC2_priority"
Flags: X - disabled, I - invalid; D - dynamic
 0    name="PC2_priority" target=10.2.1.2/32 rate=0bps/0bps total-rate=0bps packet-rate=0/0
      total-packet-rate=0 queued-bytes=0/0 total-queued-bytes=0 queued-packets=0/0 total-queued-packets=0
      bytes=1142/2624 total-bytes=0 packets=15/13 total-packets=0 dropped=0/0 total-dropped=0
```

Here you will see real-time:

- rate (PC2 is getting more)
- queued (is the lower priority host queue filling up?)
- dropped packets

- priority effect values

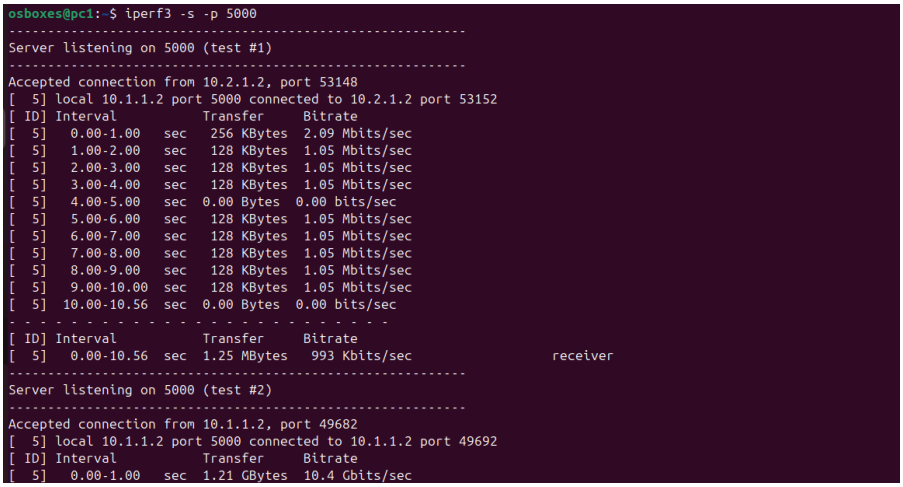
If both PC1 and PC2 send traffic, the rate in the *PC2_priority* queue is always high or drops less, then it means that the priority is working properly.

PC2 priority=1/1 - PC2 is always bandwidth ahead.

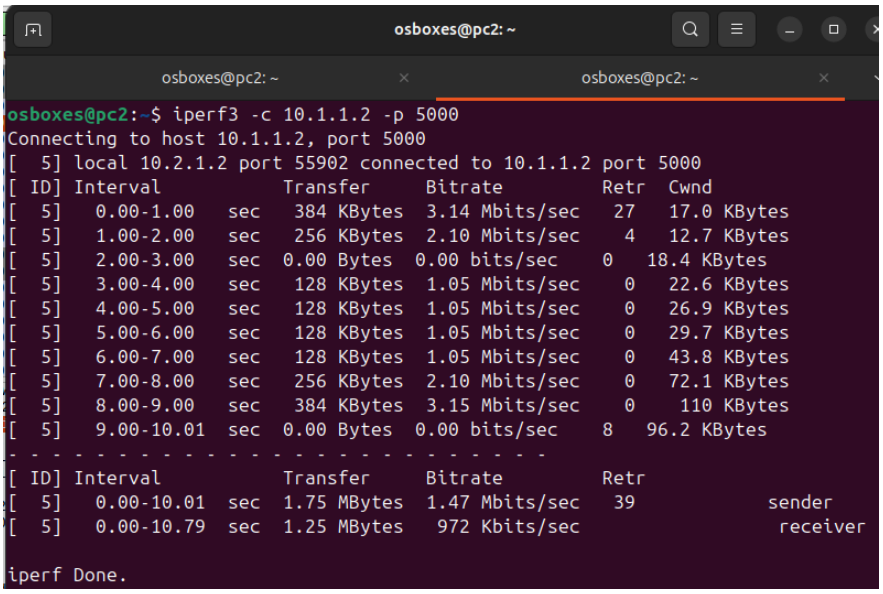
Bandwidth Test (iperf3)

Step:

PC1 (server):



PC2 (client):



First, if you test with just PC2, it shows speeds close to the max-limit (like 2 Mbps or 1 Mbps). Then run another iperf3, download from PC1 at the same time. The QoS is correct, so in case of

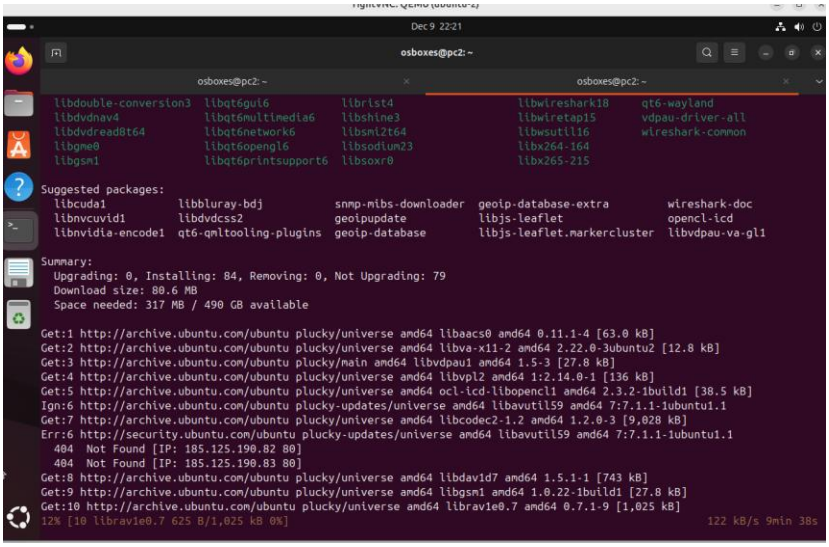
congestion, PC2 will have more throughput and PC1 will get less. This means that the priority queue is actually guaranteeing more bandwidth.

PC2 more speed than PC1 (because priority=1/1)

Package install while usingwith Priority

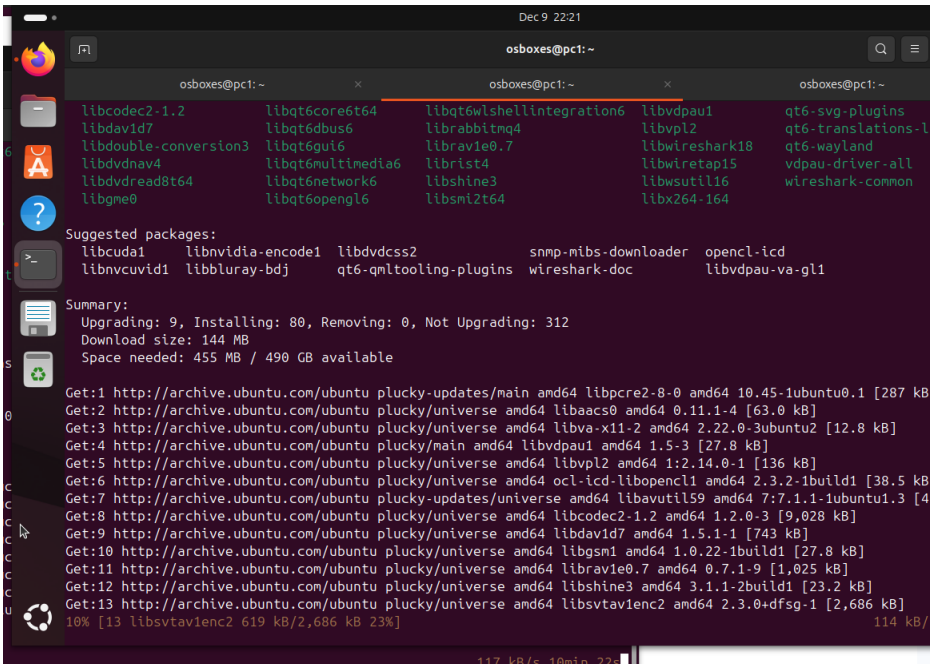
The difference in package install time is easily noticeable.

The PC2's speed is found to be higher when downloading and installing the following files:



Here, speed 122 kB/s was available for PC2.

PC1 (server):



Compare:

- PC1 speed (slow)
- PC2 speed (faster)

Priority was correct.

2. Try to use Wireshark to see the QoS packets. How does this depend on the number of routers in the network topology?

Answer:

QoS Packet Analysis with Wireshark

Wireshark was already installed and active on the PC.

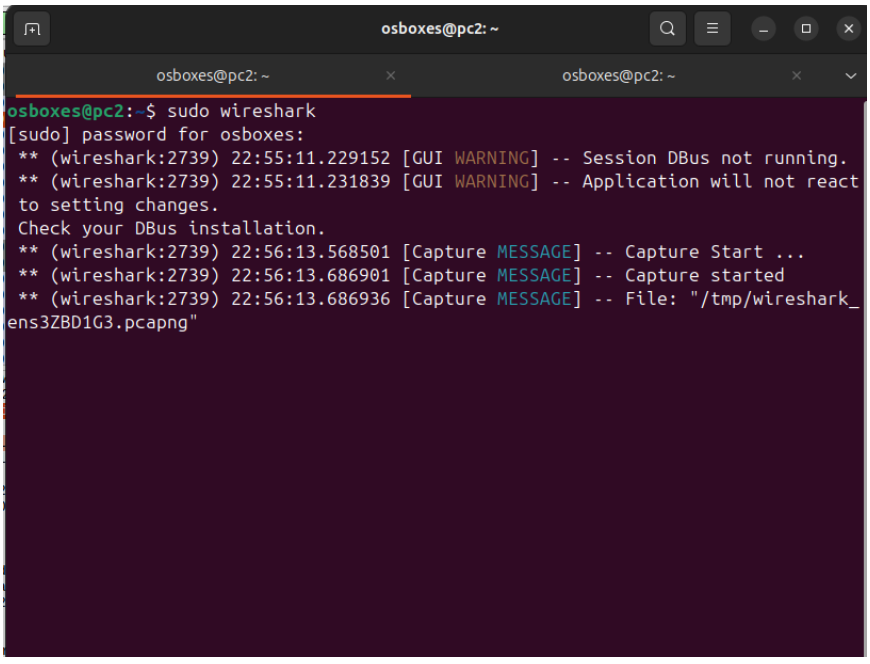
Capture interface:

- **ens3 (cloud NAT VM)**

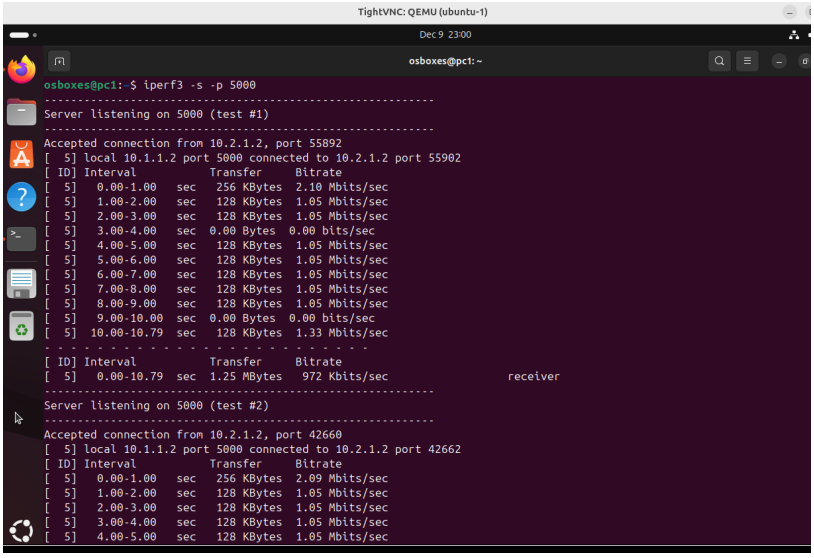
Ens3 was selected in Wireshark and the capture was started. The router to host traffic is going through this interface.

Wireshark Filter (to see QoS)

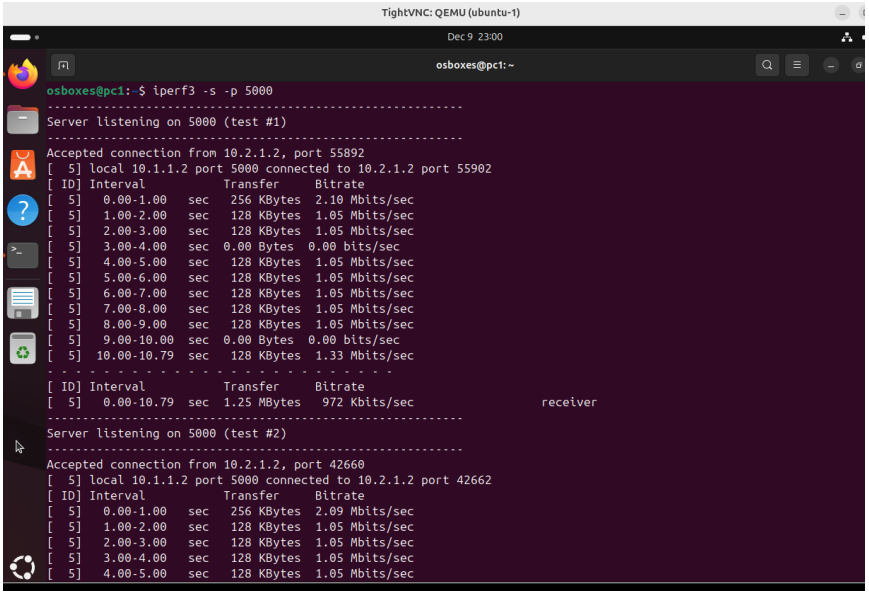
First, I entered Wireshark through PC2.



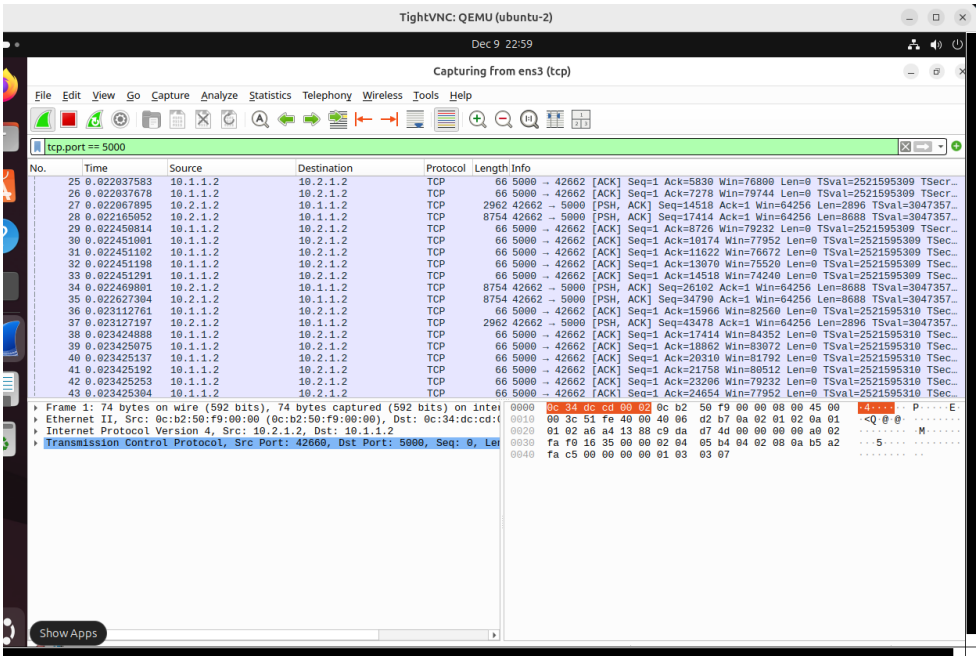
Then run the command *iperf3 -s -p 5000* in the terminal of PC1, then server listening starts.



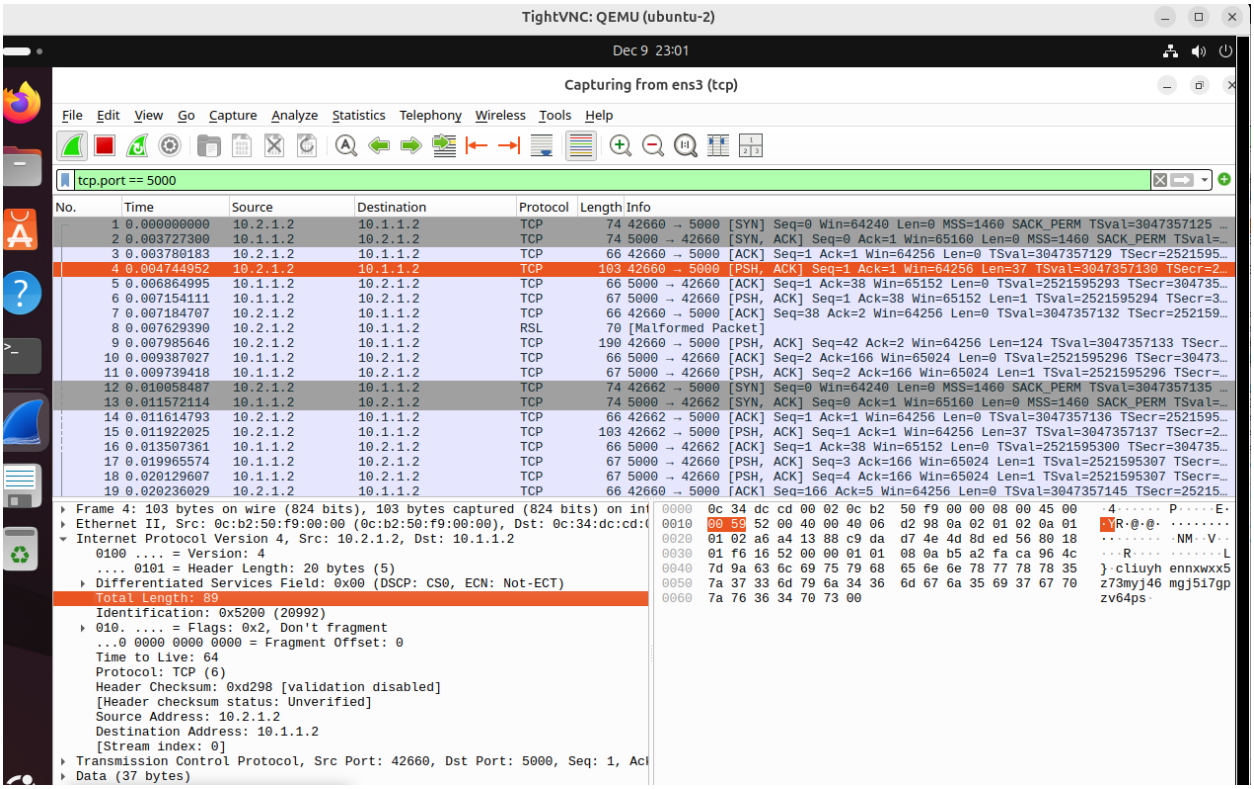
Then the file transfer starts on PC2 and the transfer starts by responding to 5000.



After entering ens3 in Wireshark, to view the packets, go to the filter option and enter *tcp.port == 5000* to view the packets and QoS packets are seen. Only the data flow of iperf3 is seen in this.



For a more detailed look, you can go to the transmission control protocol and see more details, including the type of packet.



Local QoS rules are applied which can be seen in Wireshark. If the router mark trusts, priority is preserved, otherwise it is dropped/overwritten.

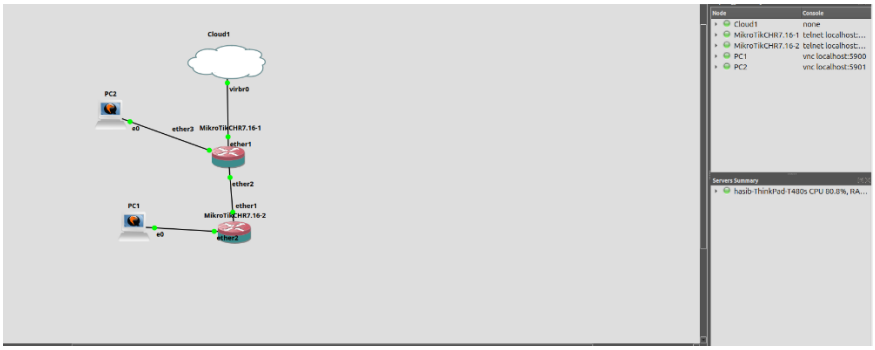
Bonus - Two routers

1. Edit your network scheme to use 2 hosts and 2 routers, each one of them has a different subnet, and the 2 hosts should be able to reach each other (direct connection between the routers and static routes is more than enough) and redo task 2.

Answer:

New Network Scheme (2 Routers + 2 Hosts + 2 Separate Subnets)

Topology:



IP ADDRESSING

Router-1 (R1)

ether1 - R2 link (towards R2):

```
[admin@MikroTik] > /ip address add interface=ether1 address=10.0.0.1/30
```

This is a point-to-point link between two routers. 10.0.0.0/30 means:

- *R1 = 10.0.0.1*
- *R2 = 10.0.0.2*
- *Network = 10.0.0.0*
- *Broadcast = 10.0.0.3*

ether2 - PC1 LAN:

```
[admin@MikroTik] > /ip address add interface=ether2 address=192.168.1.1/24
```

The R1 LAN subnet is *192.168.1.0/24* and PC1 uses this interface.

Router–2 (R2)

ether2 - Link to R1

```
[admin@MikroTik] > /ip address add interface=ether2 address=10.0.0.2/30
```

ether3 - PC2 LAN

```
[admin@MikroTik] > /ip address add interface=ether3 address=192.168.2.1/24
```

ether1 - Internet (Cloud) DHCP

```
[admin@MikroTik] > /ip dhcp-client add interface=ether1 use-peer-dns=yes add-def
ault-route=yes
```

R2 gets internet through Cloud - virbr0 - NAT - R2.

PC IP SETTINGS

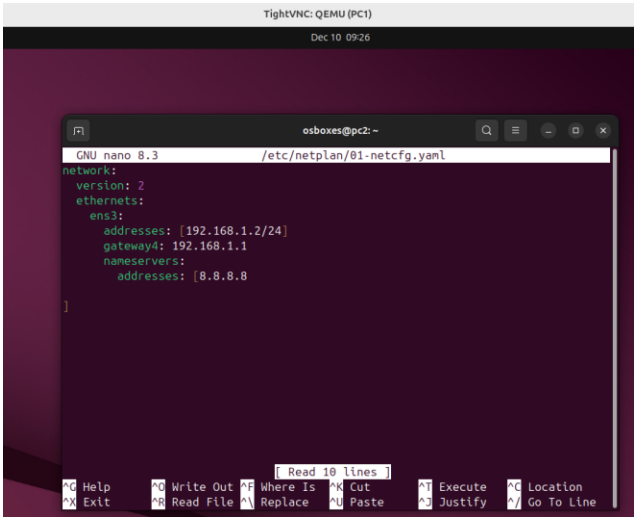
PC1

IP: 192.168.1.2

Mask: 255.255.255.0

Gateway: 192.168.1.1

DNS: 8.8.8.8



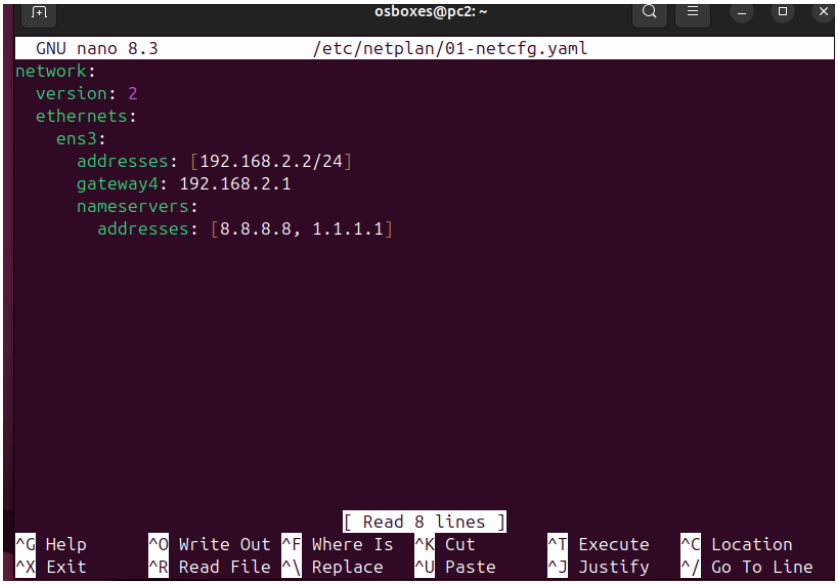
PC2

IP: 192.168.2.2

Mask: 255.255.255.0

Gateway: 192.168.2.1

DNS: 8.8.8.8



STATIC ROUTES

R1 - LAN2 route

```
[admin@MikroTik] > /ip route add dst-address=192.168.2.0/24 gateway=10.0.0.2
```

To go to PC2's network, R1 should send traffic to R2 as the next-hop.

R2 - LAN1 route

```
[admin@MikroTik] > /ip route add dst-address=192.168.1.0/24 gateway=10.0.0.1
```

By running these two commands, PC1 - PC2 communicate.

Test Connectivity

R2 internet check

```
[admin@MikroTik] > /ping 8.8.8.8
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 8.8.8.8                                    56 108 43ms894us
    1 8.8.8.8                                    56 108 42ms932us
    2 8.8.8.8                                    56 108 46ms15us
    3 8.8.8.8                                    56 108 45ms198us
    4 8.8.8.8                                    56 108 44ms556us
    sent=5 received=5 packet-loss=0% min-rtt=42ms932us avg-rtt=44ms519us
   max-rtt=46ms15us
```

The ping was successful, which means the connection was ok.

2. Do you need to write QoS rules for each router? Is there a way to let the other routers know that this packet has more priority over the other packets

Answer:

PC1 Limit (R1)

```
[admin@MikroTik] > /queue simple add name=PC1_limit target=192.168.1.2/32 max-li
mit=1M/1M
```

R1 - Give Priority to PC2

```
[admin@MikroTik] > /queue simple add name=PC2_priority target=192.168.2.2/32 max
-limit=2M/2M priority=1/1
```

R2 detects the packet and sends priority traffic to the fast queue.

Queue Monitor

Simple print stats can be used to monitor:

```
[admin@MikroTik] > /queue simple print stats
Flags: X - disabled, I - invalid; D - dynamic
 0    name="PC1_limit" target=192.168.1.2/32 rate=0bps/0bps total-rate=0bps
      packet-rate=0/0 total-packet-rate=0 queued-bytes=0/0
      total-queued-bytes=0 queued-packets=0/0 total-queued-packets=0
      bytes=0/0 total-bytes=0 packets=0/0 total-packets=0 dropped=0/0
      total-dropped=0

 1    name="PC2_priority" target=192.168.2.2/32 rate=0bps/0bps total-rate=0bps
      packet-rate=0/0 total-packet-rate=0 queued-bytes=0/0
      total-queued-bytes=0 queued-packets=0/0 total-queued-packets=0
      bytes=0/0 total-bytes=0 packets=0/0 total-packets=0 dropped=0/0
      total-dropped=0
```

R1 - Limit PC1

```
[admin@MikroTik] > /queue simple print stats where name="PC1_limit"
Flags: X - disabled, I - invalid; D - dynamic
 0    name="PC1_limit" target=192.168.1.2/32 rate=0bps/0bps total-rate=0bps
      packet-rate=0/0 total-packet-rate=0 queued-bytes=0/0
      total-queued-bytes=0 queued-packets=0/0 total-queued-packets=0
      bytes=0/0 total-bytes=0 packets=0/0 total-packets=0 dropped=0/0
      total-dropped=0
```

R1 - Give Priority to PC2

```
[admin@MikroTik] > /queue simple print stats where name="PC2_priority"
Flags: X - disabled, I - invalid; D - dynamic
 1    name="PC2_priority" target=192.168.2.2/32 rate=0bps/0bps total-rate=0bps
      packet-rate=0/0 total-packet-rate=0 queued-bytes=0/0
      total-queued-bytes=0 queued-packets=0/0 total-queued-packets=0
      bytes=0/0 total-bytes=0 packets=0/0 total-packets=0 dropped=0/0
      total-dropped=0
```

It shows:

- Rate
- queued packets
- dropped packets
- total data
- Priority queue always gets more bandwidth and less drops.

Bandwidth Test (iperf3)

PC1 server:

```
osboxes@pc1:~$ iperf3 -s -p 5000
-----------------------------------------------------------
Server listening on 5000 (test #1)
-----------------------------------------------------------
Accepted connection from 192.168.2.2, port 52582
[  5] local 192.168.1.2 port 5000 connected to 192.168.2.2 port 52594
[ ID] Interval           Transfer     Bitrate
[  5]   0.00-1.00   sec  0.00 Bytes  0.00 bits/sec
[  5]   1.00-2.00   sec   128 KBytes  1.05 Mbits/sec
[  5]   2.00-3.00   sec   128 KBytes  1.05 Mbits/sec
[  5]   3.00-4.00   sec   128 KBytes  1.05 Mbits/sec
[  5]   4.00-5.00   sec   128 KBytes  1.05 Mbits/sec
[  5]   5.00-6.00   sec  0.00 Bytes  0.00 bits/sec
[  5]   6.00-7.00   sec   128 KBytes  1.05 Mbits/sec
[  5]   7.00-8.00   sec   128 KBytes  1.05 Mbits/sec
[  5]   8.00-9.00   sec   128 KBytes  1.05 Mbits/sec
[  5]   9.00-10.00  sec   128 KBytes  1.05 Mbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate
[  5]   0.00-10.06  sec  1.00 MBytes   834 Kbits/sec                  receiver
-----------------------------------------------------------
Server listening on 5000 (test #2)
```

PC2 client:

```
osboxes@pc2:~$ iperf3 -c 192.168.1.2 -p 5000
Connecting to host 192.168.1.2, port 5000
[  5] local 192.168.2.2 port 52594 connected to 192.168.1.2 port 5000
[ ID] Interval           Transfer     Bitrate         Retr  Cwnd
[  5]   0.00-1.00   sec   256 KBytes  2.09 Mbits/sec   16   12.7 KBytes
[  5]   1.00-2.00   sec   128 KBytes  1.05 Mbits/sec    1   11.3 KBytes
[  5]   2.00-3.00   sec  0.00 Bytes  0.00 bits/sec    2   11.3 KBytes
[  5]   3.00-4.00   sec   128 KBytes  1.05 Mbits/sec    0   15.6 KBytes
[  5]   4.00-5.00   sec   128 KBytes  1.05 Mbits/sec    1   14.1 KBytes
[  5]   5.00-6.00   sec  0.00 Bytes  0.00 bits/sec    0   17.0 KBytes
[  5]   6.00-7.00   sec   128 KBytes  1.05 Mbits/sec    0   22.6 KBytes
[  5]   7.00-8.00   sec   128 KBytes  1.05 Mbits/sec    2   18.4 KBytes
[  5]   8.00-9.00   sec   128 KBytes  1.05 Mbits/sec    1   18.4 KBytes
[  5]   9.00-10.00  sec   128 KBytes  1.05 Mbits/sec    1   14.1 KBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Retr
[  5]   0.00-10.00  sec  1.12 MBytes   944 Kbits/sec   24             sender
[  5]   0.00-10.06  sec  1.00 MBytes   834 Kbits/sec                  receiver

iperf Done.
```

PC1 limit = 1 Mbps

PC2 priority = 2 Mbps - more stable throughput

Test result shows:

- PC1 stuck around 1 Mbps
- PC2 stable and higher priority

File Transfer Test

Creating a test file on PC1

```
osboxes@pc1:~$ dd if=/dev/zero of=file.bin bs=1M count=20
20+0 records in
20+0 records out
20971520 bytes (21 MB, 20 MiB) copied, 0.0272201 s, 770 MB/s
```

Sending via SCP

PC2 to PC1:

```
osboxes@pc1:~$ scp file.bin osboxes@192.168.2.2:/tmp
The authenticity of host '192.168.2.2 (192.168.2.2)' can't be established.
ED25519 key fingerprint is SHA256:QxHXhu+XPJBVpE45mpHqAsE+bXTG+CeJGXrv1SyidFY.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.2.2' (ED25519) to the list of known hosts.
osboxes@192.168.2.2's password:
file.bin                                                      1%  255KB   0.0KB/s - stalled -
file.bin                                                     97%   19MB 437.5KB/s   00:01 ETA
```

Check if QoS Rules are applied:

Check on R1:

```
[admin@MikroTik] > /queue simple print stats
Flags: X - disabled, I - invalid; D - dynamic
 0    name="PC2_priority" target=10.2.1.2/32 rate=0bps/0bps total-rate=0bps
      packet-rate=0/0 total-packet-rate=0 queued-bytes=0/0
      total-queued-bytes=0 queued-packets=0/0 total-queued-packets=0
      bytes=0/0 total-bytes=0 packets=0/0 total-packets=0 dropped=0/0
      total-dropped=0

 1    name="PC1-limit" target=10.1.1.2/32 rate=0bps/0bps total-rate=0bps
      packet-rate=0/0 total-packet-rate=0 queued-bytes=0/0
      total-queued-bytes=0 queued-packets=0/0 total-queued-packets=0
      bytes=0/0 total-bytes=0 packets=0/0 total-packets=0 dropped=0/0
      total-dropped=0
```

Values are visible:

- Rate
- Dropped
- Queued
- packet count

It means that QoS is working.

Verify with Wireshark

Wireshark initialization

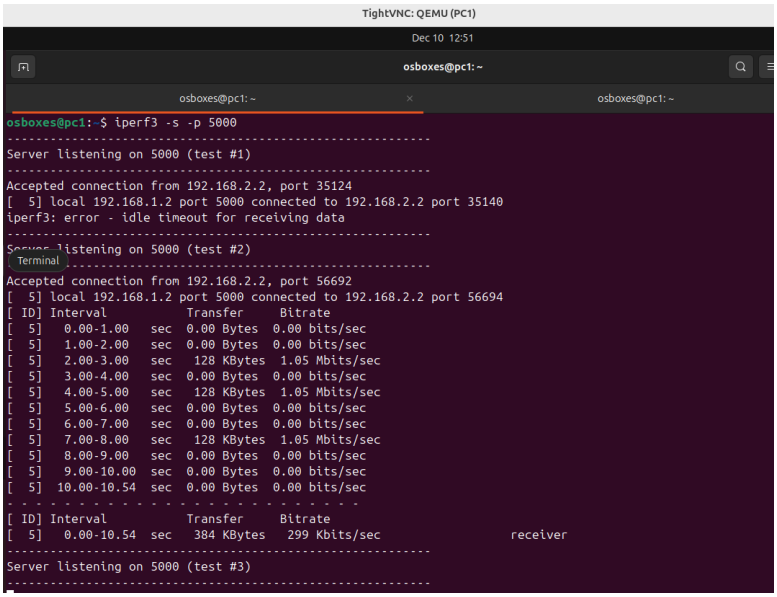
Wireshark permissions:

```
osboxes@pc2:~$ sudo usermod -aG wireshark $USER
```

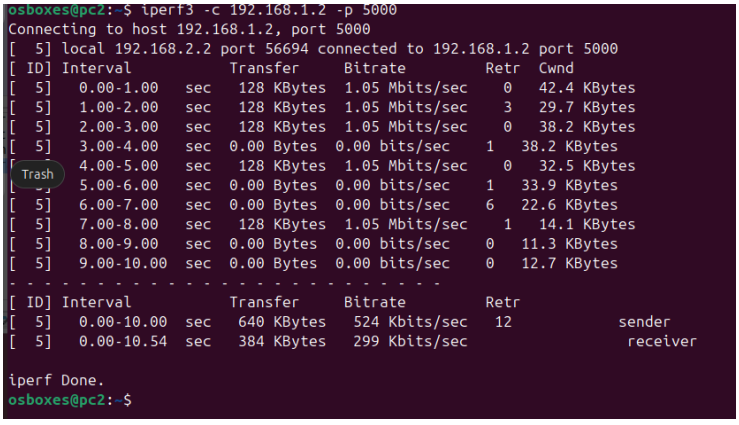
Start capturing on PC2

Interface: ens3

Then run this command iperf3 -s -p 5000 in the terminal of PC1, then server listening starts.



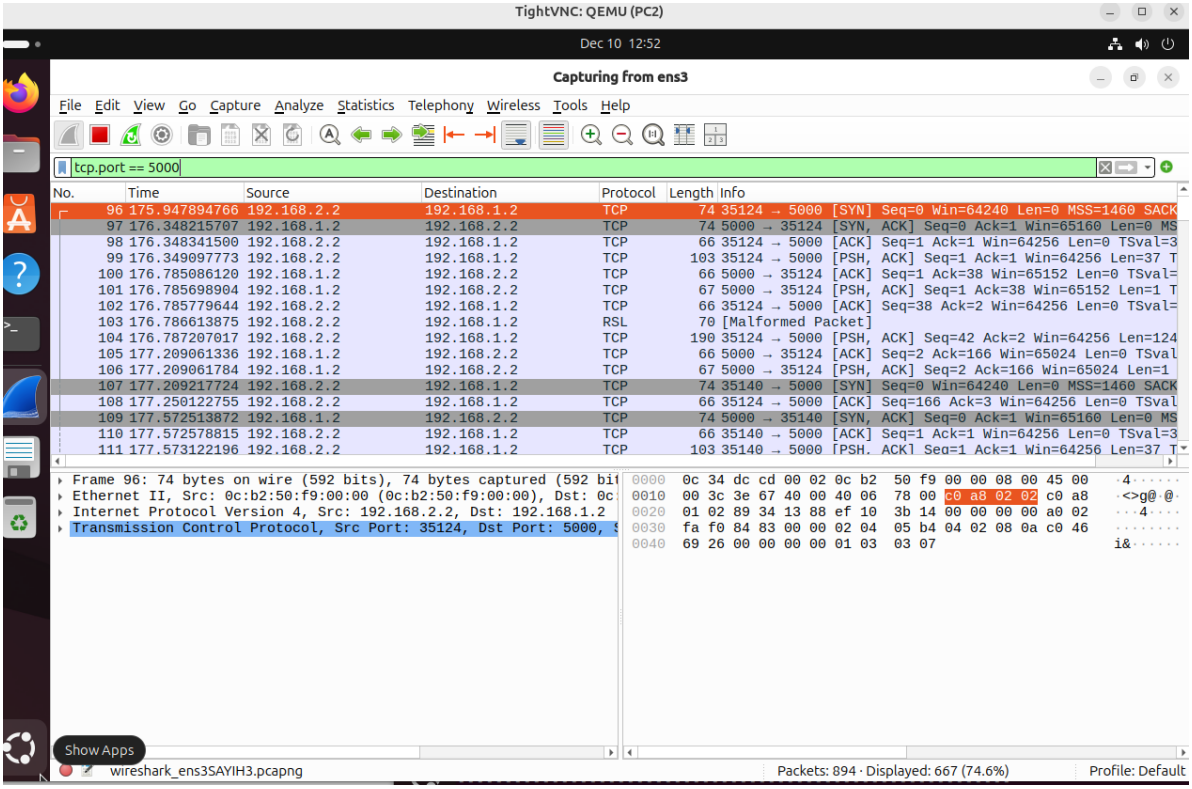
Then the file transfer starts on PC2 and the transfer starts by responding to 5000.



The interface selected was ens3.

Filter:

tcp.port == 5000



iperf3 file transfer

PC1 - iperf3 server

PC2 - file copy

Observe QoS behavior

Wireshark shows:

- PC2 stream is more stable due to congestion
- More retransmissions on low priority stream
- Ordered sequence faster

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[9] RFC 791, “Internet Protocol,” Internet Engineering Task Force, Sept. 1981. Available: https://datatracker.ietf.org/doc/html/rfc791